

IMY772 Workshop Part 1

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1)Functional Requirements

FR 1: Addition

- FR 1.1: The system should be able to add two valid hexadecimal numbers.
- FR 1.2: The system should be able add a negative number to a positive one.
- FR 1.3: The system should be able to return the original number when adding zero.

FR 2: Subtraction

- FR 2.1: The system should subtract one hexadecimal number from another.
- FR 2.2: The system should return the original number when subtracting zero.
- FR 2.3: The system should return 0 if subtraction results in a negative value.

FR 3: Multiplication

- FR 3.1: The system should be able to multiply two valid hexadecimal numbers.
- FR 3.2: The system should return 0 when multiplying by 0.
- FR 3.3: The system should return the original number when multiplying by one.

FR 4: Division

- FR 4.1: The system should divide one hexadecimal number by another.
- FR 4.2: The system should truncate decimal values resulting from division.
- FR 4.3: The system should return an error when division by zero is attempted.

FR 5: Operation Behaviour

- FR 5.1: The system should support chaining multiple operations in sequence.
- FR 5.2: For commutative operations (addition and multiplication), the system returns the same result regardless of the order in which the values are entered.
- FR 5.3: For non-commutative operations (subtraction and division), the system follows the order that the values are entered in, since swapping them changes the result.

FR 6: Input Validation

- FR 6.1: Only valid hexadecimal characters are accepted as input, digits 0–9 and letters A–F.

- FR 6.2: Each operand is limited to a maximum of two hexadecimal digits.
- FR 6.3: Any attempt to enter a non-hexadecimal character is not allowed by the system.

FR 7: Output Validation*

- FR 7.1: The system should restrict output to a maximum of 4 hexadecimal digits.
- FR 7.2: The system should not return any negative values.
- FR 7.3: The system should not return any values containing decimal places.
- FR 7.4: The system should only return valid hexadecimal output.

FR 8: Button-Based Interface

- FR 8.1: The system should provide individual buttons for each valid hexadecimal, digits 0–9 and letters A–F.
- FR 8.2: The system should provide individual buttons for each arithmetic operator (+, −, ×, ÷).
- FR 8.3: The system should be fully operable without keyboard input.

FR 9: Display

- FR 9.1: The system should display the current operand or result on screen as input is entered.
- FR 9.2: The system should display the current ongoing expression (e.g., A3 + 1B) while it is being built.
- FR 9.3: The system should update the display immediately upon each button press.

FR 10: Equals & Answer

- FR 10.1: The system should provide an equals button that triggers the evaluation of the current expression.
- FR 10.2: The system should display the computed result after the equals button is pressed.
- FR 10.3: The system should allow the result to be used as the first operand in a subsequent operation.

FR 11: Clear Operations

- FR 11.1: The system should provide a clear-all button (AC) that resets the entire calculator state.
- FR 11.2: The system should provide a clear-entry button (CE) that removes only the most recently entered digit or operand.

FR 12: Calculation History

- FR 12.1: The system should store the last 5 completed calculations.
- FR 12.2: The system should display the stored calculation history to the user on request.
- FR 12.3: The system should store each history entry as a full expression and its result (e.g. A3 + 1B = BE).

2) Category Table

Category	Functional requirements
Arithmetic Operations	FR 1 (Addition), FR 2 (Subtraction), FR 3 (Multiplication), FR 4 (Division), FR 5 (Operation Behaviour - chaining, commutativity)
Input Validation	FR 6.1 (Valid hex characters), FR 6.2 (2-digit max), FR 6.3 (Reject invalid characters)
Output Validation	FR 7.1 (4-digit max), FR 7.2 (No negatives), FR 7.3 (No decimals), FR 7.4 (Valid hex output)
Button-Based Interface	FR 8.1 (Hex digit buttons), FR 8.2 (Operator buttons), FR 8.3 (No keyboard required), FR 10.1 (Equals button), FR 11.1 (Clear all button), FR 11.2 (Clear entry button)
Display	FR 9.1 (Current operand), FR 9.2 (Ongoing expression), FR 9.3 (Immediate update), FR 10.2 (Show result)
Deletion	FR 11.2 (Clear entry), FR 11.1 (Clear all)
History	FR 12.1 (Store last 5), FR 12.2 (Display history), FR 12.3 (Store full expression)
Result Handling	FR 10.1 (Equals button), FR 10.2 (Show result), FR 10.3 (Chain from result)

Category	Functional Requirements
Arithmetic Operations	FR 1: Addition (FR 1.1–FR 1.3) – Perform hexadecimal addition including handling zero and mixed sign values. FR 2: Subtraction (FR 2.1–FR 2.3) – Perform subtraction and return 0 when the result would be negative. FR 3: Multiplication (FR 3.1–FR 3.3) – Perform multiplication including handling multiplication by 0 and 1. FR 4: Division (FR 4.1–FR 4.3) – Perform division, truncate decimal results, and prevent division by zero. FR 5: Operation Behaviour (FR 5.1–FR 5.3) – Support chained operations and correctly handle commutative and non-commutative arithmetic rules.
Input Validation	FR 6.1: Accept only valid hexadecimal characters (0–9, A–F). FR 6.2: Restrict each operand to a maximum of two hexadecimal digits. FR 6.3: Reject any non-hexadecimal characters.
Output Validation	FR 7.1: Restrict output to a maximum of four hexadecimal

	digits. FR 7.2: Prevent negative output values. FR 7.3: Prevent decimal values from being returned. FR 7.4: Ensure all outputs are valid hexadecimal values.
Button-Based Interface	FR 8.1: Provide buttons for each hexadecimal digit (0–9, A–F). FR 8.2: Provide buttons for arithmetic operators (+, −, ×, ÷). FR 8.3: Ensure the calculator is fully operable using buttons without keyboard input. FR 10.1: Provide an equals button that evaluates the current expression. FR 11.1: Provide a clear-all button (AC) that resets the entire calculator state. FR 11.2: Provide a clear-entry button (CE) that removes only the most recently entered digit or operand.
Display	FR 9.1: Display the current operand or result as input is entered. FR 9.2: Display the ongoing expression while it is being constructed. FR 9.3: Update the display immediately after each button press. FR 10.2: Display the computed result after evaluation.
Clear/ Deletion Operations	FR 11.1: Provide a clear-all button (AC) that resets the entire calculator state. FR 11.2: Provide a clear-entry button (CE) that removes only the most recently entered digit or operand.
Calculation History	FR 12.1: Store the last five completed calculations. FR 12.2: Allow the user to view stored calculation history. FR 12.3: Store each history entry as the full expression and result (e.g., $A3 + 1B = BE$).
Result Handling	FR 10.1: Provide an equals button that evaluates the current expression. FR 10.2: Display the computed result after evaluation. FR 10.3: Allow the computed result to be reused as the first operand in the next calculation.

